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A Novel Motor Imagery EEG Decoding Method Integrating Cascaded Common Spatial Pattern and Riemannian Geometry

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***Abstract*—Brain computer interfaces (BCIs) provide a pathway between the human brain and extrinsic equipment which have been widely used in rehabilitation robots. Disabled people can directly command BCI-based robots to finish various tasks just through thinking. In BCI systems, EEG decoding is a crucial part directly affecting the effectiveness of the assistance function. Aiming at improving EEG decoding performance, a novel EEG classification method combining multidomain features is proposed. Specifically, an integrated framework based on cascaded common spatial pattern and Riemannian Geometry is designed to refine spatial features. Besides, a feature fusion strategy based on filter bank, Hilbert transform and this integrated framework is effectively used to extract frequency-time-spatial features for classification. The effectiveness of this proposed method is validated on dataset BCI IV 2a. The average accuracy is 85.6% that outperforms four typical methods. Therefore, our work provides distinct advantages over conventional approaches which contributes to the further development of BCIs.**

I. Introduction

Brain computer interfaces (BCIs) provide effective ways for people to operate external devices by converting brain signals to computer commands [1]. To be specific, when subjects imagine different actions of body parts, brain signals are recorded [2]. Then BCI systems can recognize specific actions and provide valuable control information. Electroencephalography (EEG) is one of the most investigated brain signals due to its non-invasiveness, high portability, low power and accurate timing resolution [3]. Various EEG application paradigms have been proposed, among which motor imagery (MI) is a widespread paradigm in BCIs. MI can reduce physical burden and have fine control efficiency. However, MI is often accompanied by low signal-to-noise ratio, and obvious disturbance [4]. Therefore, how to extract optimal features in a low-SNR environment is a critical step which are related to the accuracy of EEG decoding and further impacts the interaction performance of BCI systems [5]. To improve EEG decoding ability, numerous feature extraction methods based on time, spatial, and frequency domain information [6] have been proposed including Common Spatial Pattern (CSP), Riemannian Geometry, Fourier Transform and Phase Space Reconstruction. In the past, researchers have employed CSP and many CSP variants, such as regularized CSP (R-CSP) [7], correlation-based CSP (CCSP) [8], Tikhonov regularization CSP (TRCSP) [9], filter bank CSP (FBCSP) [10], common spatio-spectral pattern (CSSP) [11] and sub-band CSP (SBCSP) [12] for EEG decoding. These approaches have demonstrated the effectiveness in EEG decoding, however, the core principle of CSP is to extract valuable features through projection which imposes limitations on further advancements in EEG decoding performance.

MI-based EEG classification methods mainly include deep learning (DL) and machine learning (ML). Mane et al. [13] proposed FBCNet to classify MI-based EEG signals. Schirrmeister et al. [14] introduced two DL framework, i.e., DeepConvNet and ShallowConvNet to classify MI-based EEG signals. Although these methods exhibit good performance, these advanced DL methods require extensive data computation and are suboptimal for processing nonlinear data of small and medium scale [15]. ML methods are more effective for high-dimensional data with small sample sizes, and exhibit strong generalization capabilities for unseen data and greater flexibility in decision-making. In the ML field, SVM and LDA are the most popular classifiers for EEG decoding which have been widely used and achieve excellent classification results. Shyu et al. [16] proposed a novel SVM approach that leverages the proximity of data points to the SVM decision boundary. Yang et al. [17] introduced a novel LDA classifier with an adaptive decision surface to handle EEG signals in MI tasks. These ML methods effectively classify EEG signals with reduced complexity compared to DL methods.

In this paper, to overcomes the limitations of CSP-based methods, we propose a novel EEG decoding method that combines CSP with Riemannian Geometry. First, raw EEG signals are fed into multiple sub-bands and a single broadband, respectively. Secondly, the phase information of filtered signal is obtained using Hilbert Transform, which enhances the potential to improve EEG decoding performance. Furthermore, all calculated EEG information including amplitude and phase angle passes through an integrated framework utilizing cascaded CSP and Riemannian geometry, and are aggregated into fusion feature vectors for classification. Finally, the classification model is trained and test performance is validated by using SVM on BCI IV 2a dataset respectively. The major contributions are two-fold:1) An integrated framework based on cascaded common spatial pattern and Riemannian tangent space (CCSP-RTS) is

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proposed which contributes to extract more comprehensive spatial features. 2) A feature fusion strategy utilizing filter banks, Hilbert transform and CCSP-RTS is given to obtain valuable frequency-time-spatial features.

The content of this study is arranged as follows: Section II introduces the proposed methodology in detail. Section III outlines the experimental design, with a focus on the BCI Competition IV 2a dataset. In Section IV, we present the results along with a comprehensive analysis. Section V presents the conclusion.

II. METHODOLOGY

A. Cascaded CSP

CSP approach designs a spatial filter *w* by maximizing the differences of EEG signals from different category and the similarity of EEG signals from the same category. Then the desired spatial filter W is constructed by selecting the first *m* and last *m* rows of *w* and the projected feature matrix Z can be computed by

$$Z = W^T X \tag{1}$$

where $X \in R^{N_C \times N_S}$, N_C and N_S are the number of electrodes and samples, respectively.

To enhance classification performance, cascaded CSP embedding a hierarchical multi-layer structure is introduced to capture deeper and more meaningful features [18]. For the *j*-th sub-band and the *k*-th cascaded layer, the corresponding spatial filter and projected feature matrix are represented as W_k^j and Z_k^j, respectively. The combined feature matrices from all sub-bands can be expressed as

$$Z_k = [Z_k^1, Z_k^2, \cdots, Z_k^n] \tag{2}$$

$$Z_k^j = (W_k^j)^T Z_{k-1}^j \tag{3}$$

When $k = 1$, $Z_{k-1}^j = X^j$ denotes the EEG signal of the *j*-th sub-band.

B. Riemannian Geometry

In BCI systems, Riemannian distance-based methods are commonly used to extract features from EEG covariance matrices located in the Riemannian manifold. The Riemannian distance represents the shortest path between two points in a curve way, which can be calculated by

$$\delta_R(M_1, M_2) = \| \log(M_1^{-1} M_2) \|_F \tag{4}$$

where $\|\cdot\|_F$ is Frobenius norm, and $\log(\cdot)$ is logarithmic operation, M_1 and M_2 are spatial covariance matrices, can be numerically represented as follows

$$M_i = \frac{1}{L-1} X X^T \tag{5}$$

EEG spatial covariance matrix M_i can be projected from the Riemannian manifold onto the tangent space via logarithmic mapping in (6), and mapped back to the manifold using exponential mapping in (7), as shown in Fig. 1.

$$T_i = Log_{\overline{M}}(M_i) = \overline{M}^{\frac{1}{2}} Log(\overline{M}^{-\frac{1}{2}} M_i \overline{M}^{-\frac{1}{2}}) \overline{M}^{\frac{1}{2}} \tag{6}$$

$$M_i = Exp_{\overline{M}}(T_i) = \overline{M}^{\frac{1}{2}} Exp(\overline{M}^{-\frac{1}{2}} T_i \overline{M}^{-\frac{1}{2}}) \overline{M}^{\frac{1}{2}} \tag{7}$$

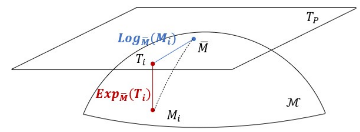


Figure 1. Riemannian Manifold and Riemannian Tangent Space

As expressed in (6) and (7), the spatial covariance matrix M_i is mapped to Riemannian tangent space at the Riemannian mean matrix $\overline{M}$. In order to calculate Riemannian mean [15], we need to find M that minimizes the sum of squared Riemannian distances from all samples, expressed in the following form.

$$\overline{M} = \mathfrak{M}(M_1, M_2, \cdots, M_N) = \arg\min_{M} \sum_{i=1}^{N} \delta_R^2(M, M_i) \tag{8}$$

To obtain Riemannian tangent space features, the upper triangle elements of spatial covariance matrices are selected.

$$f_v = vect(upper(\overline{M}^{-\frac{1}{2}} Log_{\overline{M}}(M_i) \overline{M}^{-\frac{1}{2}})) \in R^{(Nc+1)Nc/2} \tag{9}$$

where f_v is vectorized features of *v-th* trial, *upper (·)* functions to extract the upper triangular part of the matrix and *vect (·)* functions to vectorize non-diagonal elements of the matrix.

C. Hilbert Transform

The EEG phase angle has extensive potential information which is helpful for improving classification accuracy. Hilbert transform [19] is broadly used to obtain the phase information by

$$\hat{x}(t) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{x(\tau)}{\tau - t} d\tau \tag{10}$$

$$X(t) = [x_1(t), x_2(t), \ldots, x_{N_C}(t)]^T \tag{11}$$

The analytic signal $\Psi(t)$ and corresponding phase angle $\varphi(t)$ are defined as follows

$$\Psi(t) = x(t) + j\hat{x}(t) \tag{12}$$

$$\varphi(t) = \arctan \frac{\hat{x}(t)}{x(t)} \tag{13}$$

D. Feature Fusion Strategy

In this study, we proposed an integrated framework combined cascaded CSP and Riemannian Geometry to implement comprehensive refinement on features, as shown in Fig. 2. The first part cascaded CSP aims to extract spatial feature matrices Z_i with strong representativeness by performing multiple projections. This is helpful for improving classification accuracy. Then we project the spatial covariance matrices of projected outputs Z_i to Riemannian tangent space, from which we can obtain the Riemannian distance-based features that capture more accurate correlation between EEG signals.

Based on this integrated framework, a feature fusion strategy is given to refine both sub-band features and broadband features. The detailed feature fusion strategy is shown in Algorithm 1 and Fig. 2.

In step 3 of Algorithm 1, joint amplitude and phase signals $\{X^j, \varphi^j\} \in R^{N_C \times N_S}$ contain more discriminative information. During Steps 4–6, joint EEG signals are condensed and refiltered by two cascaded CSPs which enhance feature discriminability and reduce redundant information. Steps 7-10 project all CSP-based feature matrices to the Riemannian tangent space to refine the fusion features from different perspective which contain affluent frequency-time-spatial domain information. Finally, the linear kernel-based SVM is chose for our MI classification tasks.

III. EXPERIMENTS

A. Dataset Description

In this study, we use BCI IV 2a dataset to verify the effectiveness of proposed framework. The BCI IV 2a dataset was collected from 9 healthy subjects with 22 EEG channels and 3 EOG channels. The specific MI tasks include right hand, left hand, both feet and tongue. In our work, we only focused on left-hand and right-hand MI signals on 22 EEG channels during session 1, which includes 144 trials with 72 trials per MI task. Fig. 3 illustrates the timing sequence paradigm of trials. At the onset, an auditory beep was presented, accompanied by the display of a fixation cross for 2 s. After that, a directional cue in the form of an arrow pointing left, right, up and down appeared for 1.25 s, during which subjects were instructed to perform specific MI for 3 s [20].

B. Experimental Setting

During our experiment, 1000 epoch samples from the start of the cue are selected. Therefore, each trial had N_C = 22 EEG channels, $N_S = 1000$ epoch samples, $N_i = 2$ classes MI tasks. Besides, we apply band-pass filtering to segment the EEG signals into 17 overlapping frequency bands (4 Hz bandwidth, 2 Hz overlap), ranging from 4–8 Hz up to 36–40 Hz, as well as a broad 4–40 Hz band.

Two-layer cascaded CSP were designed for the binary classification tasks. Based on experimental experience, the number of components m_1, m_2 were selected as 4 and 3 to obtain optimal spatial filters W_1, W_2 of cascaded CSP.

Additionally, a 5-fold cross-validation procedure is utilized to mitigate overfitting and statistical bias during classification. Comprehensive comparisons with typical feature extraction methods are given to validate the performance of proposed framework.

(1) CSP

(2) Riemannian Geometry

(3) FBCSP

(4) SBCCSP [18]

(5) CCSP-RTS

The classification approach employs a SVM with a linear kernel to classify the given dataset. The penalty parameter is set to C=1, while other hyperparameters retain their default values.

IV. RESULTS AND ANALYSIS

A. Comparison with Other Methods

In our integrated framework, the number of spatial filter components 2m is an important factor that significantly affects classification performance. To determine the optimal spatial filters applied in cascaded CSP, we compare the classification performance under spatial filter components $m_1, m_2 = \{2,1\}, \{4,3\}, \{6,5\}, \{8,7\}, \{10,9\}$ which has shown in Fig. 4. Both accuracy and kappa value initially increase and then decline with the number of components increasing. Therefore, we select $m_1 = 4$ and $m_2 = 3$ as the default parameters for our experiment.

To intuitively compare extracted features of different methods, we apply t-SNE [21] to reduce dimensionality and visualize the feature distributions of two subjects. As is shown in Fig. 5, CCSP-RTS method yields more discriminative features for subjects 01 and 05, outperforming the typical CSP and RTS methods. Therefore, our proposed method has a relative advantage in binary classification tasks.

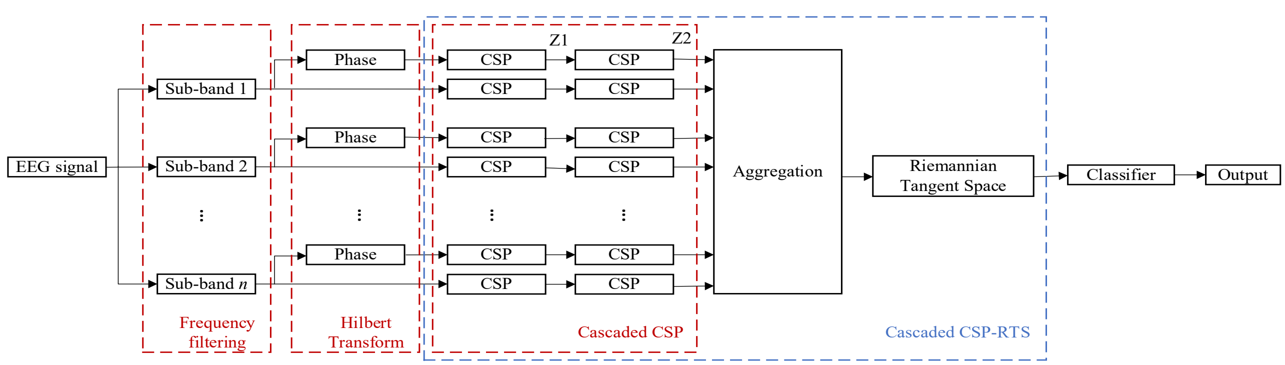


Figure 2. Overall framework of EEG decodin

Algorithm 1: Sub-bands CCSP-RTS based Feature Fusion

Input: N trials of EEG signal $X \in R^{N_C \times N_S}$ from class $i \in \{1,2\}$

Output: features $\widetilde{F}$

1: for $j = 1; j \le n; j++$ do

2: filter X into signal X^j of frequency band j;

3: Calculate phase angle φ^j of X^j by Hilbert Transform and obtain joint signals $\{X^j, \varphi^j\} \in R^{N_C \times N_S}$;

4: Calculate the spatial filters $\{W_1^j, \hat{W}_1^j\} \in R^{2m_1 \times N_C}$ corresponding to the first layer of cascaded CSP and filter $\{X^j, \varphi^j\}$ to get the projected matrices $\{Z_1^j, \hat{Z}_1^j\} \in R^{2m_1 \times N_S}$;

5: Calculate the spatial filters $\{W_2^j, \hat{W}_2^j\} \in R^{2m_2 \times 2m_1}$ corresponding to the second layer of cascaded CSP and filter $\{Z_1^j, \hat{Z}_1^j\}$ to get the projected matrices $\{Z_2^j, \hat{Z}_2^j\} \in R^{2m_2 \times N_S}$;

6: end

7: Aggregate $\{Z_2^j, \hat{Z}_2^j\}$ from all sub-bands into $Z_2 = [Z_2^1, Z_2^2, \cdots, Z_2^n] \in R^{n \times 2m_2 \times N_S}$, $\hat{Z}_2 = [\hat{Z}_2^1, \hat{Z}_2^2, \cdots, \hat{Z}_2^n] \in R^{n \times 2m_2 \times N_S}$ and reshape into matrices $\{D_2, \hat{D}_2\} \in R^{N_{CAS} \times N_S}$ with $N_{CAS} = n \times 2m_2$;

8: Generate the spatial covariance matrices $\{M_v, \hat{M}_v\} \in R^{N_{CAS} \times N_{CAS}}$ based on $\{D_2, \hat{D}_2\}$ for trial $v \in \{1, 2, \cdots, N\}$;

9: Calculate the Riemannian mean $\{\overline{M}, \overline{\hat{M}}\} \in R^{N_{CAS} \times N_{CAS}}$ between N covariance matrices;

10: Obtain features $\{f_v, \hat{f}_v\} \in R^{(N_{CAS}+1)N_{CAS}/2}$ from Riemannian tangent space;

11: concatenate features $F = [f_1, f_2, \cdots, f_N]^T \in R^{N \times (N_{CAS}+1)N_{CAS}/2}$ and $\hat{F} = [\hat{f}_1, \hat{f}_2, \cdots, \hat{f}_N]^T \in R^{N \times (N_{CAS}+1)N_{CAS}/2}$ into $\widetilde{F} \in R^{N \times (N_{CAS}+1)N_{CAS}}$;

12: return the fusion features $\widetilde{F}$

The classification performance of five methods is presented in Table I and II. Compared with other methods, our proposed method demonstrates superior accuracy. It achieves an average accuracy of 85.6%, with improvements of 11.1%, 9.3%, 3%, and 2.8% over the other methods, respectively. The accuracy improvement is particularly obvious for subjects 01 and 05 with accuracy rates of 92.4% and 91.6%, respectively. Nevertheless, for subjects 03 and 08, there is still room for accuracy amelioration in our method. Besides, the two proposed methods exhibit standard deviations of 12.5% and 11.7%, respectively—both lower than those of the other four methods, as shown in Fig. 6. This indicates that our methods have better stability and robustness. Furthermore, our proposed method achieves kappa index of 0.71 which is higher than all other methods. The results indicate our method is more reliable. Notably, for subjects 03, 07, and 08, the kappa values surpass 0.85, demonstrating a strong agreement between the predicted results and actual values. Therefore, our proposed method delivers better classification performance and outperforms other methods.

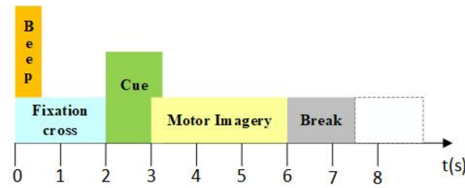


Figure 3. The timing sequence of BCI competition IV dataset 2a

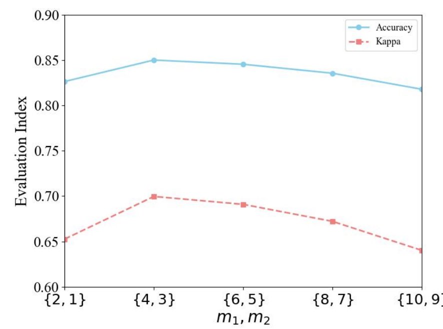


Figure 4. Classification performance of CCSP-RTS under different spatial filter components m_1, m_2

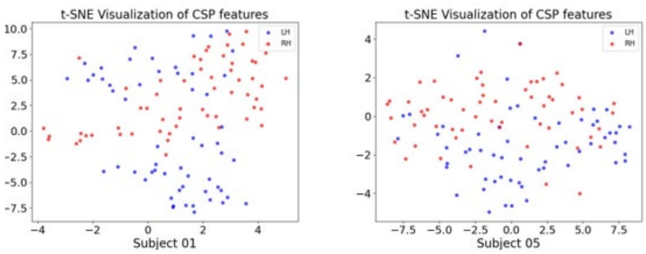
B. Phase Information Analysis

The phase angle φ can enhance the classification performance due to more potential information. To validate the effectiveness of additional phase information, we compare classification results under two cases which are features extracted from X and $\{X, \varphi\}$ respectively. As is shown in Table I and II, the second case which takes both amplitude and phase features into consideration has higher average accuracy and kappa value. The majority of subjects show improved classification accuracy and kappa value. Besides, Fig. 6 shows latter case has lower standard deviation. These results illustrate the advantage of introducing phase information in our framework.

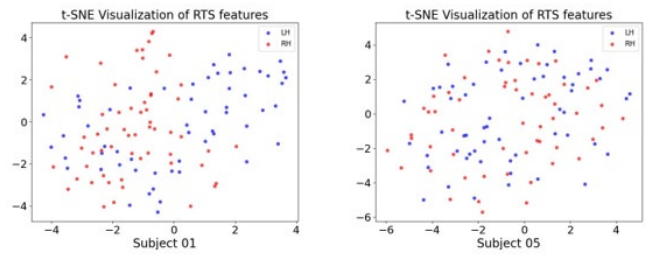
C. Discussion

Feature extraction is a critical challenge for BCI systems, as it directly impacts the effectiveness in assisting disabled individuals. The traditional CSP algorithm demonstrates good performance in MI-EEG classification. However, this method suffers from poor generalization performance and is highly susceptible to noise interference. Riemannian geometry is increasingly used to extract more robust features for classification, utilizing covariance matrices to capture valuable spatial features instead of relying on spatial filters.

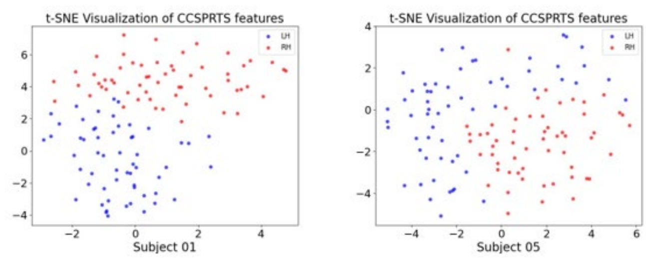
In our work, we proposed a CCSP-RTS method to make full use of the merits of CSP and Riemannian geometry and extract more distinguished features. The CCSP-RTS method combines cascaded CSP and Riemannian manifold, outperforming SBCCSP, which demonstrates the importance of introducing Riemannian geometry. Besides, incorporating phase information to extract valuable features further enhances the classification accuracy.



(a) CSP feature distributions



(b) RTS feature distributions



(c) CCSP-RTS feature distributions

Figure 5. Feature distributions of Subjects 01 and 05 with t-SNE, where LH: left hand, RH: right hand.

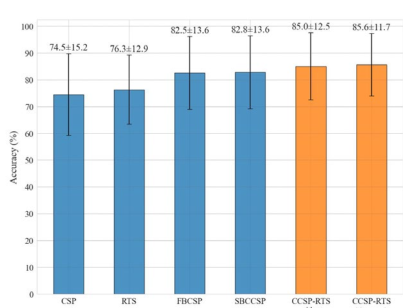


Figure 6. Classification performance of different methods .

Although our proposed method achieved good classification performance, there are several aspects that should be further improved. First, the selection of frequency bandwidth is an important aspect to investigate which influences both classification performance and training time. Besides, there has a relatively high dimension of features due to multiple frequency bands and added phase information. Dimensionality reduction while improving classification accuracy is a crucial problem, which will be proposed in our future research. The CCSP-RTS method presents a novel feature extraction architecture in the field of machine learning which offers distinct advantages. With deep-learning method developing, how to apply the proposed method to neural networks is also a question worth considering.

TABLE I. 5-FOLD CROSS-VALIDATION CLASSIFICATION ACCURACY(%) OF SVM

Subject	Method					
	CSP	***RTS***	***FBCSP***	***SBCCSP***	***CCSP-RTS with*** X	***CCSP-RTS with*** $\{X, \varphi\}$
S01	77.1	79.1	90.3	90.3	91.0	92.4
S02	59.7	64.7	59.8	61.8	59.7	63.9
S03	95.8	94.4	95.1	93.0	96.5	95.8
S04	67.3	77.1	76.4	68.7	78.4	78.4
S05	52.8	57.6	90.2	90.2	91.6	91.6
S06	67.3	68.8	69.4	67.4	72.2	71.5
S07	75.6	75.7	95.1	94.4	93.0	93.0
S08	98.6	97.2	95.8	98.6	96.5	97.2
S09	76.4	72.2	70.8	80.5	86.1	86.8
Average	74.5	76.3	82.6	82.8	85.0	85.6

TABLE II. 5-FOLD CROSS-VALIDATION CLASSIFICATION KAPPA VALUE OF SVM

Subject	Method					
	CSP	***RTS***	***FBCSP***	***SBCCSP***	***CCSP-RTS with*** X	***CCSP-RTS with*** $\{X, \varphi\}$
S01	0.54	0.58	0.8	0.81	0.82	0.85
S02	0.21	0.3	0.2	0.26	0.21	0.29
S03	0.92	0.89	0.9	0.86	0.93	0.92
S04	0.34	0.53	0.52	0.38	0.56	0.56
S05	0.05	0.15	0.8	0.8	0.83	0.83
S06	0.33	0.34	0.38	0.35	0.44	0.43
S07	0.51	0.51	0.9	0.88	0.86	0.86
S08	0.97	0.94	0.91	0.97	0.93	0.94
S09	0.53	0.45	0.41	0.61	0.72	0.73
Average	0.49	0.52	0.65	0.66	0.7	0.71

V. Conclusion

In this paper, a novel EEG decoding method combining frequency-time-spatial features is proposed. We obtain valuable frequency and time information based on filter bank and Hilbert transform. Then an integrated framework of cascaded CSP and Riemannian geometry is proposed to extract critical features from EEG signal further. The fusion strategy combining filter bank, Hilbert transform and our proposed integrated framework is given for multidomain feature extraction. Afterward, fusion features are then classified by a linear kernel-based SVM. The feasibility of proposed method is evaluated on BCI Competition IV Dataset 2a and the experimental results show better classification performance with average accuracy of 85.6%. In the future, we will focus more on EEG decoding and related control for BCI-based robots to help disabled people.

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